

Department of Networks and Information Security

Network Administration (10686213)

Total Credits 3

major compulsory

Prerequisites P1 : Computer Networks (68134)

Course Contents

This course deals with network and system administration principles. It mainly concentrates on the basic troubleshooting techniques for Operation System (OS) booting- related problems, network failures, and security issues. Topics to be covered in this course include UNIX-based server administration, shell scripting basics, user management, file management, network file sharing, networked printers, Dynamic Host Configuration Protocol (DHCP) and Domain Name Service (DNS) troubleshooting, web and mail servers administration. Finally, security-related administration will be introduced.

Intended Learning Outcomes (ILO's)	Student Outcomes (SO's)	Contribution
1 Understand basic administration concepts for both system and network components.	A	15 %
2 Getting familiar with Linux operating system as a platform to apply different administrative tasks.	B	20 %
3 Automating administrative tasks related to processes, users, files, and periodic processes management.	C	25 %
4 Exploiting the gained system administration concepts to effective network administration, including basic configurations, subnetting, and service management.	K	25 %
5 Improve communication skills through doing oral discussion.	F	5%
6 Improve team working through conducting a project.	D, J	10%

Textbook and/ or References

UNIX and LINUX System Administration Handbook, 4th Edition.

Author(s): Evi Nemeth, Grath Synder, Trent Hein, and Ben Whaley; ISBN-13: 978-0131480056

Assessment Criteria	Percent (%)
Midterm Exam	30 %
Labs	10 %
Project	25 %
Final Exam	35 %

Course Plan

Week	Topic
1	· Introduction to Network Administration.
1-2	· Introduction to UNIX System Administration.
3_4	· Scripting and the Shell.
5	· Process Management
6	· File Management.
6	· Midterm Exam
6-7	· Periodic Process & Log Files
7	· User Management
8-10	· TCP/IP Network Basics and Subnetting.
11-12	· Linux Networking Basics (Interfaces, DNS, Services, and DHCP).
13	· Network File System (NFS)
14	· Security Administration
15-16	· Virtualization + Final Exam

